Biological Modelling Assignmment 3

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October 2026

1 Question 1

Given the predator–prey model;

$$\frac{dN_1}{dt} = N_1(a_1 - b_{11}N_1 - b_{12}N_2)$$

$$\frac{dN_2}{dt} = N_2(-a_2 + b_{21}N_1)$$

a) Obtain the equilibrium points of the system.

b) Obtain the Jacobian matrix and hence determine the nature and stability of the fixed points.

c) Simulate the predator–prey model using values of a_1, b_{11}, b_{12}, a_2, and b_{21} from existing literature

a) Equilibrium Points

At equilibrium, there is no rate of change at any given time in both populations so set both differential equations equal to zero:

$$\frac{dN_1}{dt} = 0, \qquad \frac{dN_2}{dt} = 0.$$

Hence,

$$N_1(a_1 - b_{11}N_1 - b_{12}N_2) = 0$$

$$N_2(-a_2 + b_{21}N_1) = 0.$$

We assume that all parameters $a_1, a_2, b_{11}, b_{12}, b_{21}$ are positive and are a representation of the intrinsic rate of growth, the carrying capacity and the predation coefficient.

Find points at equilibrium

$$N_1 = 0 \quad or \quad a_1 - b_{11}N_1 - b_{12}N_2 = 0.$$

Similarly, Equation (2) gives

$$N_2 = 0 \quad or \quad -a_2 + b_{21}N_1 = 0.$$

Finding the Equilibrium Points

At equilibrium, both differential equations are equal to zero.

From Equation (1):

$$N_1(a_1 - b_{11}N_1 - b_{12}N_2) = 0$$

$$N_1 = 0$$

or

$$a_1 - b_{11}N_1 - b_{12}N_2 = 0.$$

From Equation (2):

$$N_2(-a_2 + b_{21}N_1) = 0$$

$$N_2 = 0$$

or

$$-a_2 + b_{21}N_1 = 0.$$

$$b_{21}N_1 = a_2$$

$$N_1 = \frac{a_2}{b_{21}}.$$

We consider the four possible combinations of these conditions.

Case 1: $N_1 = 0$ **and** $N_2 = 0$

Substituting $N_1 = 0$ and $N_2 = 0$ into Equations (1) and (2) gives

$$0(a_1 - b_{11}(0) - b_{12}(0)) = 0$$

and

$$0(-a_2 + b_{21}(0)) = 0.$$

Both equations are satisfied. Therefore, the first equilibrium point is

$$O(0, 0)$$

.

Case 2: $N_2 = 0$ **and** $a_1 - b_{11}N_1 - b_{12}N_2 = 0$

Substituting $N_2 = 0$ into the second condition gives

$$a_1 - b_{11}N_1 - b_{12}(0) = 0.$$

$$a_1 - b_{11}N_1 = 0.$$

$$b_{11}N_1 = a_1.$$

$$N_1 = \frac{a_1}{b_{11}}.$$

Since $N_2 = 0$, the second equilibrium point is

$$A = \left(\frac{a_1}{b_{11}}, 0\right).$$

Case 3: $N_1 = 0$ **and** $-a_2 + b_{21}N_1 = 0$

Substituting $N_1 = 0$ into the second condition gives

$$-a_2 + b_{21}(0) = 0.$$

$$-a_2 = 0 \quad \Longrightarrow \quad a_2 = 0.$$

However, $a_2 > 0$ by assumption. This is a contradiction. Therefore, this combination does not produce an equilibrium point.

Case 4: $a_1 - b_{11}N_1 - b_{12}N_2 = 0$ **and** $-a_2 + b_{21}N_1 = 0$

From the second condition,

$$-a_2 + b_{21}N_1 = 0.$$

$$b_{21}N_1 = a_2.$$

$$N_1 = \frac{a_2}{b_{21}}.$$

Substituting into the first condition gives

$$a_1 - b_{11}\left(\frac{a_2}{b_{21}}\right) - b_{12}N_2 = 0.$$

$$b_{12}N_2 = a_1 - \frac{b_{11}a_2}{b_{21}}.$$

$$b_{12}N_2 = \frac{a_1 b_{21} - b_{11}a_2}{b_{21}}.$$

$$N_2 = \frac{a_1 b_{21} - b_{11}a_2}{b_{21}b_{12}}.$$

The third equilibrium point is

$$B = \left(\frac{a_2}{b_{21}}, \frac{a_1 b_{21} - b_{11}a_2}{b_{21}b_{12}}\right).$$

For this equilibrium point to represent coexistence, both populations must be positive. Since $a_2, b_{21}, b_{11}, b_{12} > 0$, the condition $N_2 > 0$ requires

$$a_1 b_{21} - b_{11}a_2 > 0.$$

Thus,

$$a_1 b_{21} > b_{11}a_2,$$

$$\frac{a_1}{a_2} > \frac{b_{11}}{b_{21}}.$$

Conclusion

The system has three equilibrium points:

- $O(0,0)$: The extinction equilibrium, where both the prey and predator populations are zero.
- $A\,(a_1 b_{11}, 0)$: The prey-only equilibrium, where the prey population exist in the absence of predators.
- $B\,(a_2 b_{21}, a_1 b_{21} - b_{11}a_2 b_{21} b_{12})$: The coexistence equilibrium, which is positive provided a1 a2 ¿ b11 b21 .

b) Jacobian Matrix and Stability of the Fixed Points

The given predator–prey system is

$$\frac{dN_1}{dt} = N_1(a_1 - b_{11}N_1 - b_{12}N_2)$$

$$\frac{dN_2}{dt} = N_2(-a_2 + b_{21}N_1).$$

Let

$$f(N_1, N_2) = N_1(a_1 - b_{11}N_1 - b_{12}N_2)$$

and

$$g(N_1, N_2) = N_2(-a_2 + b_{21}N_1).$$

The Jacobian matrix is defined as

$$J(N_1, N_2) = \begin{bmatrix} \frac{\partial f}{\partial N_1} & \frac{\partial f}{\partial N_2} \\ \frac{\partial g}{\partial N_1} & \frac{\partial g}{\partial N_2} \end{bmatrix}$$

Calculate the Partial Derivatives

First function

$$f(N_1, N_2) = a_1N_1 - b_{11}N_1^2 - b_{12}N_1N_2.$$

Differentiate with respect to N_1,

$$\frac{\partial f}{\partial N_1} = a_1 - 2b_{11}N_1 - b_{12}N_2.$$

Differentiate with respect to N_2,

$$\frac{\partial f}{\partial N_2} = -b_{12}N_1.$$

Second function

$$g(N_1, N_2) = -a_2N_2 + b_{21}N_1N_2.$$

Differentiate with respect to N_1,

$$\frac{\partial g}{\partial N_1} = b_{21}N_2.$$

Differentiate with respect to N_2,

$$\frac{\partial g}{\partial N_2} = -a_2 + b_{21}N_1.$$

Therefore, the Jacobian matrix is

$$J(N_1, N_2) = \begin{bmatrix} a_1 - 2b_{11}N_1 - b_{12}N_2 & -b_{12}N_1 \\ b_{21}N_2 & -a_2 + b_{21}N_1 \end{bmatrix}$$

Check for stability of $O(0,0)$

Substituting $N_1 = 0$ and $N_2 = 0$ into the Jacobian matrix

$$J(0,0) = \begin{bmatrix} a_1 & 0 \\ 0 & -a_2 \end{bmatrix}$$

Calculate the eigenvalues

$$\det(J - \lambda I) = 0.$$

$$\begin{vmatrix} a_1 - \lambda & 0 \\ 0 & -a_2 - \lambda \end{vmatrix} = 0$$

$$(a_1 - \lambda)(-a_2 - \lambda) = 0.$$

$$\lambda_1 = a_1, \qquad \lambda_2 = -a_2.$$

Since $a_1 > 0$ and $a_2 > 0$,

$$\lambda_1 > 0, \qquad \lambda_2 < 0.$$

The eigenvalues have opposite signs therefore

$$O = (0,0)$$

is an unstable saddle point.

Stability of $A\left(\frac{a_1}{b_{11}}, 0\right)$

Substituting

$$N_1 = \frac{a_1}{b_{11}}, \qquad N_2 = 0$$

into the Jacobian matrix

$$J(A) = \begin{bmatrix} a_1 - 2b_{11}\left(\frac{a_1}{b_{11}}\right) & -b_{12}\left(\frac{a_1}{b_{11}}\right) \\ b_{21}(0) & -a_2 + b_{21}\left(\frac{a_1}{b_{11}}\right) \end{bmatrix}$$

$$J(A) = \begin{bmatrix} -a_1 & -\frac{a_1 b_{12}}{b_{11}} \\ 0 & -a_2 + \frac{a_1 b_{21}}{b_{11}} \end{bmatrix}$$

Since this is an upper triangular matrix, its eigenvalues are the diagonal entries:

$$\lambda_1 = -a_1$$

and

$$\lambda_2 = -a_2 + \frac{a_1 b_{21}}{b_{11}}.$$

$$\lambda_2 = \frac{a_1 b_{21} - b_{11} a_2}{b_{11}}.$$

Since $\lambda_1 < 0$, the stability of A depends on the sign of λ_2.
If

$$\frac{a_1}{a_2} < \frac{b_{11}}{b_{21}},$$

then $\lambda_2 < 0$. Both eigenvalues are negative making it a locally asymptotically stable node.
If

$$\frac{a_1}{a_2} > \frac{b_{11}}{b_{21}},$$

then $\lambda_2 > 0$. The eigenvalues have opposite signs and making it an unstable saddle point
If

$$\frac{a_1}{a_2} = \frac{b_{11}}{b_{21}},$$

then $\lambda_2 = 0$. The equilibrium is non-hyperbolic, so the linearisation alone cannot determine its stability. At this equality, the coexistence point B coincides with A.

Stability of the Coexistence Equilibrium B

$$B = \left(\frac{a_2}{b_{21}}, \frac{a_1 b_{21} - b_{11} a_2}{b_{21} b_{12}} \right).$$

Substituting B into the Jacobian gives

$$J(B) = \begin{bmatrix} a_1 - 2b_{11}\left(\frac{a_2}{b_{21}}\right) - b_{12}\left(\frac{a_1 b_{21} - b_{11} a_2}{b_{21} b_{12}}\right) & -b_{12}\left(\frac{a_2}{b_{21}}\right) \\ b_{21}\left(\frac{a_1 b_{21} - b_{11} a_2}{b_{21} b_{12}}\right) & -a_2 + b_{21}\left(\frac{a_2}{b_{21}}\right) \end{bmatrix}.$$

$$J(B) = \begin{bmatrix} -\frac{b_{11}a_2}{b_{21}} & -\frac{b_{12}a_2}{b_{21}} \\ \frac{a_1b_{21}-b_{11}a_2}{b_{12}} & 0 \end{bmatrix}.$$

Obtaine the eigen values

$$\det(J(B) - \lambda I) = 0.$$

$$\begin{vmatrix} -\frac{b_{11}a_2}{b_{21}} - \lambda & -\frac{b_{12}a_2}{b_{21}} \\ \frac{a_1b_{21}-b_{11}a_2}{b_{12}} & -\lambda \end{vmatrix} = 0.$$

$$\lambda\left(\frac{b_{11}a_2}{b_{21}} + \lambda\right) + \frac{a_2(a_1b_{21} - b_{11}a_2)}{b_{21}} = 0.$$

$$\lambda^2 + \frac{b_{11}a_2}{b_{21}}\lambda + \frac{a_2(a_1b_{21} - b_{11}a_2)}{b_{21}} = 0.$$

Using the quadratic formula,

$$\lambda = \frac{-\frac{b_{11}a_2}{b_{21}} \pm \sqrt{\left(\frac{b_{11}a_2}{b_{21}}\right)^2 - \frac{4a_2(a_1b_{21}-b_{11}a_2)}{b_{21}}}}{2}.$$

The stability of B is determined by the eigenvalues. Since

$$\frac{b_{11}a_2}{b_{21}} > 0$$

and the coexistence equilibrium exists when

$$a_1b_{21} - b_{11}a_2 > 0,$$

the characteristic equation has a positive coefficient of λ and a positive constant term. Hence, both eigenvalues have negative real parts.

Therefore, the coexistence equilibrium B is globally asymptotically stable. Depending on the discriminant, it may be a stable node or a stable spiral.

c) Model simulation

The parameter values used to simulate the predator–prey model were obtained from the study by Forgan [1] which investigated predator–prey dynamics in self-replicating interstellar probes. The study provides parameter values for a logistic-growth predator–prey model. These values were adapted to the model

considered in this study by comparing the coefficients of the two systems of differential equations.

The model is given by

$$\frac{dN_1}{dt} = N_1(a_1 - b_{11}N_1 - b_{12}N_2), \tag{1}$$

$$\frac{dN_2}{dt} = N_2(-a_2 + b_{21}N_1). \tag{2}$$

The values used in the simulation are in Table 2.

The coefficient b_{11} was calculated from the prey growth rate and carrying capacity reported by Forgan [1], such that

$$b_{11} = \frac{b_R}{K_R} = \frac{0.6}{20} = 0.03.$$

Table 1: Parameter values

Parameter	**Value**	**Description**
a_1	0.6	Prey growth rate
b_{11}	0.03	Prey density-dependent limitation
b_{12}	1.333	Effect of predators on prey
a_2	1.0	Predator death rate
b_{21}	1.0	Contribution of prey to predator growth
$N_1(0)$	1.8	Initial prey population
$N_2(0)$	1.0	Initial predator population

Python code

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.integrate import solve_ivp

# Parameters
a1 = 0.6
b11 = 0.6 / 20
b12 = 1.333
a2 = 1.0
b21 = 1.0

# Initial conditions
N10 = 1.8
N20 = 1.0

# Predator-prey model
def predator_prey(t, N):
    N1, N2 = N
    dN1_dt = N1 * (a1 - b11*N1 - b12*N2)
    dN2_dt = N2 * (-a2 + b21*N1)
```

```
    return [dN1_dt, dN2_dt]

# Numerical simulation
t = np.linspace(0, 100, 5000)

solution = solve_ivp(
    predator_prey,
    [0, 100],
    [N10, N20],
    t_eval=t,
    rtol=1e-9,
    atol=1e-11
)

N1, N2 = solution.y

# Interior equilibrium
N1_eq = a2 / b21
N2_eq = (a1 - b11*N1_eq) / b12

# Plot 1: Population dynamics
plt.figure(figsize=(9, 5))
plt.plot(solution.t, N1, label="Prey population")
plt.plot(solution.t, N2, label="Predator population")
plt.xlabel("Time")
plt.ylabel("Population")
plt.title("Predator–Prey Population Dynamics")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.savefig("population_dynamics.png", dpi=300)
plt.show()

# Plot 2: Phase portrait
plt.figure(figsize=(7, 5))
plt.plot(N1, N2, label="Trajectory")
plt.scatter(
    N1_eq, N2_eq, color="red",
    label="Interior equilibrium", zorder=3
)
plt.xlabel("Prey population (N1)")
plt.ylabel("Predator population (N2)")
plt.title("Predator–Prey Phase Portrait")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.savefig("phase_portrait.png", dpi=300)
plt.show()

print(
    "Interior equilibrium:",
    round(N1_eq, 4),
    round(N2_eq, 4)
)
```

Listing 1: Python implementation of the predator–prey model

Outputs

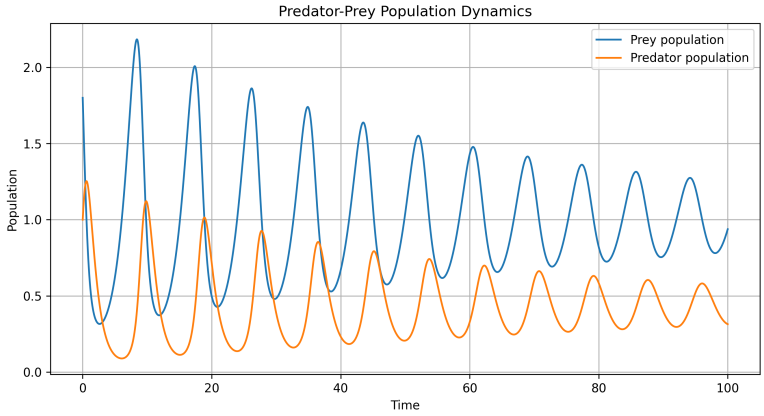


Figure 1: Predator–prey population dynamics over time.

Figure 3 shows how the prey and predator populations change over the simulation period.

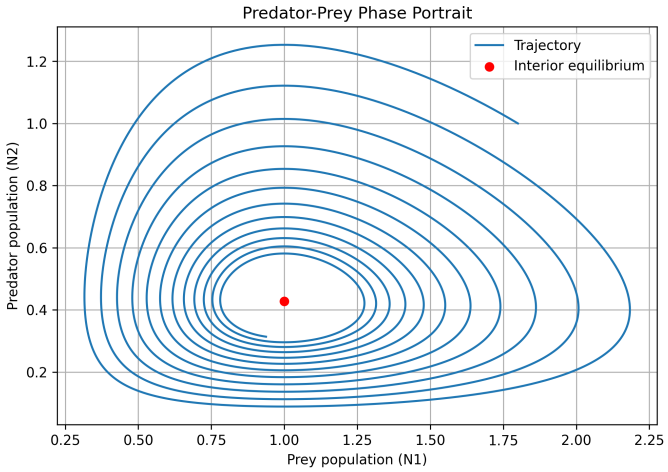


Figure 2: Phase portrait of the predator–prey model.

Figure 2 illustrates the relationship between the prey and predator populations and shows the interior equilibrium point.

2 Question 3

Simulate the model below

$$\frac{dx}{dt} = x(2 - 3x - 4y), \tag{3}$$

$$\frac{dy}{dt} = y(-3 + 5x). \tag{4}$$

Parameter Selection

The parameter values were specified in the question. The initial population values were adapted from a published simulation by Rivera-Estay et al. [2], which reports initial population values of $X(0) = 15$ and $Y(0) = 10$ for two populations in a three-species model. These values were adopted as illustrative initial conditions.

Table 2: Parameter values

Parameter	Value	Description
a_1	2	Prey growth rate
b_{11}	3	Prey density-dependent limitation
b_{12}	4	Effect of predators on prey
a_2	3	Predator death rate
b_{21}	5	Contribution of prey to predator growth
$x(0)$	15	Initial prey population
$y(0)$	10	Initial predator population

Python Code

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.integrate import solve_ivp

def model(t, population):
    x, y = population
    dxdt = x * (2 - 3*x - 4*y)
    dydt = y * (-3 + 5*x)
    return [dxdt, dydt]

# Initial conditions
x0, y0 = 1.0, 1.0

# Time interval
t_eval = np.linspace(0, 50, 1000)

# Solve the model
solution = solve_ivp(
    model, (0, 50), [x0, y0], t_eval=t_eval
```

```
)

t = solution.t
x = solution.y[0]
y = solution.y[1]

# Plot population dynamics
plt.figure(figsize=(10, 5))
plt.plot(t, x, label="Prey (x)")
plt.plot(t, y, label="Predator (y)")
plt.xlabel("Time")
plt.ylabel("Population")
plt.title("Predator–Prey Population Dynamics")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.savefig("population_dynamics.png", dpi=300)
plt.show()

# Plot phase diagram
plt.figure(figsize=(7, 6))
plt.plot(x, y, label="Trajectory")
plt.plot(x0, y0, "go", label="Initial point")
plt.plot(x[-1], y[-1], "ro", label="Final point")
plt.xlabel("Prey population (x)")
plt.ylabel("Predator population (y)")
plt.title("Phase Diagram")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.savefig("phase_diagram.png", dpi=300)
plt.show()
```

Listing 2: Python code for the predator–prey simulation

2.1 Outputs

Figure 3 shows how the prey and predator populations change over time.

Figure 4 shows the trajectory of the predator and prey populations in the phase plane.

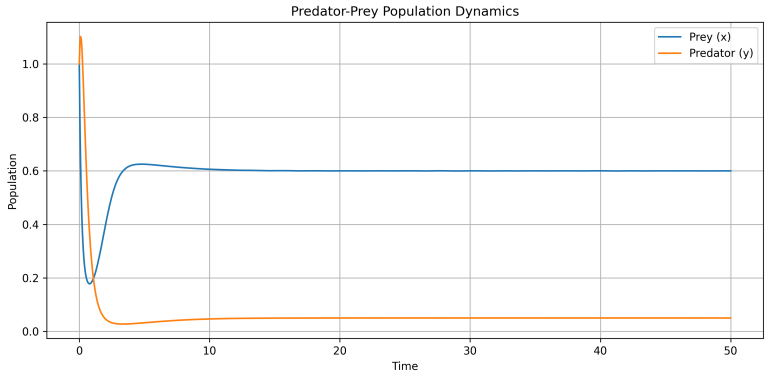


Figure 3: Prey and predator population dynamics over time.

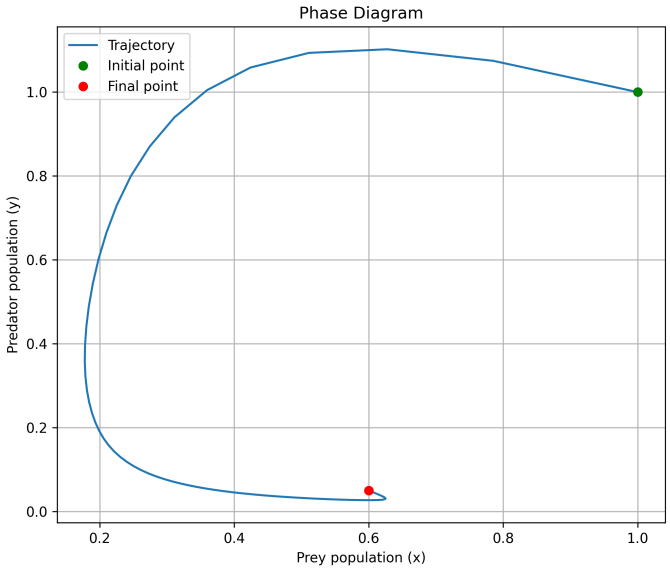


Figure 4: Phase portrait showing the population trajectory and interior equilibrium.

Question 3

Obtain all the fixed points for the facultative mutualism model given by

$$\frac{dN_1}{dt} = N_1(a_1 - b_{11}N_1 + b_{12}N_2)$$
$$\frac{dN_2}{dt} = N_2(a_2 + b_{21}N_1 - b_{22}N_2).$$

where a_j and b_{ij} are positive constants.

The two species experience intraspecific competition. In the absence of mutualistic interaction, each species grows logistically.

At equilibrium, both derivatives are equal to zero. Therefore,

$$N_1(a_1 - b_{11}N_1 + b_{12}N_2) = 0$$

$$N_2(a_2 + b_{21}N_1 - b_{22}N_2) = 0.$$

From equation 1

$$N_1 = 0 \quad or \quad a_1 - b_{11}N_1 + b_{12}N_2 = 0.$$

From equation 2

$$N_2 = 0 \quad or \quad a_2 + b_{21}N_1 - b_{22}N_2 = 0.$$

Case 1: $N_1 = 0$ **and** $N_2 = 0$

Taking both populations to be zero gives

$$O(0, 0).$$

Both populations go to extinction

Case 2: $N_2 = 0$ **and** $N_1 \neq 0$

Substituting $N_2 = 0$ into Equation (1) gives

$$N_1(a_1 - b_{11}N_1 + b_{12}(0)) = 0.$$

$$N_1(a_1 - b_{11}N_1) = 0.$$

Since $N_1 \neq 0$,

$$a_1 - b_{11}N_1 = 0.$$

$$b_{11}N_1 = a_1.$$

$$N_1 = \frac{a_1}{b_{11}}.$$

The second fixed point is

$$A = \left(\frac{a_1}{b_{11}}, 0\right).$$

Case 3: $N_1 = 0$ **and** $N_2 \neq 0$

Substituting $N_1 = 0$ into Equation (2) gives

$$N_2(a_2 + b_{21}(0) - b_{22}N_2) = 0.$$

$$N_2(a_2 - b_{22}N_2) = 0.$$

Since $N_2 \neq 0$,

$$a_2 - b_{22}N_2 = 0.$$

$$b_{22}N_2 = a_2.$$

$$N_2 = \frac{a_2}{b_{22}}.$$

The third fixed point is

$$B = \left(0, \frac{a_2}{b_{22}}\right).$$

Case 4: $N_1 \neq 0$ **and** $N_2 \neq 0$

Since neither population is zero, the expressions inside the brackets in Equations (1) and (2) must equal zero. Thus,

$$a_1 - b_{11}N_1 + b_{12}N_2 = 0$$

$$a_2 + b_{21}N_1 - b_{22}N_2 = 0.$$

From Equation (3),

$$b_{11}N_1 = a_1 + b_{12}N_2.$$

$$N_1 = \frac{a_1 + b_{12}N_2}{b_{11}}.$$

Substitute into Equation (4)

$$a_2 + b_{21}\left(\frac{a_1 + b_{12}N_2}{b_{11}}\right) - b_{22}N_2 = 0.$$

$$a_2b_{11} + a_1b_{21} + b_{21}b_{12}N_2 - b_{11}b_{22}N_2 = 0.$$

$$a_2b_{11} + a_1b_{21} = (b_{11}b_{22} - b_{21}b_{12})N_2.$$

$$N_2 = \frac{a_2b_{11} + a_1b_{21}}{b_{11}b_{22} - b_{21}b_{12}}$$

To obtain N_1, substitute Equation (6) into Equation (5):

$$N_1 = \frac{a_1 + b_{12}\left(\frac{a_2 b_{11} + a_1 b_{21}}{b_{11}b_{22} - b_{21}b_{12}}\right)}{b_{11}}.$$

$$N_1 = \frac{a_1(b_{11}b_{22} - b_{21}b_{12}) + b_{12}(a_2 b_{11} + a_1 b_{21})}{b_{11}(b_{11}b_{22} - b_{21}b_{12})}.$$

$$N_1 = \frac{a_1 b_{11}b_{22} - a_1 b_{21}b_{12} + a_2 b_{11}b_{12} + a_1 b_{21}b_{12}}{b_{11}(b_{11}b_{22} - b_{21}b_{12})}.$$

$$N_1 = \frac{a_1 b_{11}b_{22} + a_2 b_{11}b_{12}}{b_{11}(b_{11}b_{22} - b_{21}b_{12})}.$$

$$N_1 = \frac{a_1 b_{22} + a_2 b_{12}}{b_{11}b_{22} - b_{21}b_{12}}.7$$

Therefore, the coexistence fixed point is

$$C = \left(\frac{a_1 b_{22} + a_2 b_{12}}{b_{11}b_{22} - b_{21}b_{12}}, \frac{a_2 b_{11} + a_1 b_{21}}{b_{11}b_{22} - b_{21}b_{12}}\right).$$

The following are the equilibrium points;

- $O = (0, 0)$: Both species are absent. This represents the extinction equilibrium, which is not caused by mutualistic interaction.
- $A = \left(\frac{a_1}{b_{11}}, 0\right)$: This represents the equilibrium population of species N_1 in the absence of species N_2.
- $B = \left(0, \frac{a_2}{b_{22}}\right)$: This represents the equilibrium population of species N_2 in the absence of species N_1.
- $C = (N_1^*, N_2^*)$: This is the positive coexistence equilibrium, where both species coexist:

$$N_1^* = \frac{a_1 b_{22} + a_2 b_{12}}{b_{11}b_{22} - b_{21}b_{12}},$$
$$N_2^* = \frac{a_2 b_{11} + a_1 b_{21}}{b_{11}b_{22} - b_{21}b_{12}}.$$

For the coexistence equilibrium to be biologically feasible, both population values must be positive. Since all parameters are positive, this requires

$$b_{11}b_{22} - b_{21}b_{12} > 0.$$

References

[1] D. H. Forgan, "Predator-prey behaviour in self-replicating interstellar probes," *International Journal of Astrobiology*, vol. 18, no. 6, pp. 552–561, 2019.

[2] V. Rivera-Estay, F. Córdova-Lepe, F. N. Moreno-Gómez *et al.*, "Exploring the effects of competition and predation on the success of biological invasion through mathematical modeling," *Scientific Reports*, vol. 14, p. 4416, 2024.